

Consumer reactions to chatbot versus human service: An investigation in the role of outcome valence and perceived empathy

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ABSTRACT

Service outcome valence is a prominent contextual factor that has been under-researched in studies of consumer response to automated service-givers. We investigate whether consumers respond differently to chatbot and human customer service when faced with positive versus negative service outcomes. We also explore the role of perceived empathy as a potential mediator in the focal relationship. 714 individuals participated in three studies based on the vignette method. Study participants reported significantly lower satisfaction, repatronage intentions, recommendation acceptance, and recommending the provider to others following interactions with a chatbot compared with a human agent in both positive and negative service outcome conditions. The effect was fully mediated by the service-giver's perceived empathy. Increasing a chatbot's perceived empathy via more empathetic communication (but not human-like appearance) improved consumer evaluations of chatbot service relative to a less empathetic chatbot configuration and matched evaluations of the human agent. A fourth study involved a quasi-experiment in which 100 individuals interacted with ChatGPT to obtain medical advice. This study corroborated our conclusions about the impact of perceived empathy on consumers' service experience and provider ratings.

1. Introduction

Chatbots have become ubiquitous in customer service encounters online and on the phone. Although, until recently, companies have used chatbots primarily in supporting roles to human agents. Chatbots normally handle basic inquiries while routing complex cases to human agents. Now, the advent of generative artificial intelligence (AI) programs, such as ChatGPT and Gemini, is finally setting organizations on a path toward full automation of customer service provision.

Customer support via AI-powered chatbots provides essential benefits relative to human-agent service. Chatbot service is more cost-efficient, available on demand, highly scalable, consistent, and less susceptible to factors that degrade human service, such as stress and fatigue.

Hence, companies have a strong incentive to replace human customer service agents with AI chatbots. At the same time, consumers show mixed receptivity toward chatbot-based customer service. On one hand, there is general excitement about generative AI. On the other

hand, generative AI chatbots are controversial, and concerns persist about their ethics, trustworthiness, and propensity to hallucinate or spread misinformation (e.g., see [AI Incident Database \(2024\)](#) for a current catalog of problems caused by AI). Industry studies consistently show that while some consumers favor chatbot customer service, a large majority of participants (e.g., 88% in the [Ipsos \(2023\)](#) study) prefer human agents in most contexts.

Although consumer acceptance of chatbot service will likely evolve with improvements in technology, one's response to automation is also influenced by various person-specific and contextual factors which are independent of technology (e.g., [Rese et al., 2020](#); also, see [Lapointe and Rivard, 2005](#) for a model of consumer resistance). One ubiquitous contextual factor that has received no attention in research on consumer response to automated service-givers involves service outcome valence—whether an outcome was good or bad for the customer. Wide-ranging research in psychology and marketing shows that goal attainment and outcome valence influence how individuals experience events, feel, and behave (e.g., [Brady et al., 2006](#); [Kaur et al., 2022](#);

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Lourenço et al., 2022).

Therefore, this paper makes a contribution to the literature by empirically examining how consumers perceive their service experience when served by a chatbot versus a human agent in encounters that produce divergent—either positive or negative—outcomes. We elucidate the mechanism accounting for the observed results and evaluate three approaches to help improve consumer perceptions of chatbot service. We provide evidence of an empathy-based process for evaluating human and chatbot service agents. Our research is further distinguished by using commercially available generative AI to produce some of the evidence.

In what follows, we present four studies conducted in three distinct service contexts. Study 1 was used to establish participants' baseline response to positive service outcomes enabled either by an AI chatbot or a human agent. Study 2 built on Study 1 and involved a negative service outcome for the consumer (but, of note, not a service process failure). We used data from Studies 1 and 2 to test for mediation between our key constructs. Study 3 extended these findings to consider ways of boosting consumer evaluations of chatbot service. Study 4 entailed a quasi-experiment in which individuals interacted with an actual generative AI chatbot, ChatGPT, to obtain medical advice. This study was conducted to produce corroborating evidence for the observed impacts of perceived provider empathy on customer outcomes in a field-experiment-type setting.

2. Theory and hypotheses

2.1. Consumer preference for customer service providers: human vs. chatbot

Consumer preference for new technologies is strongly driven by their perceived capabilities (Lutfi et al., 2023). Consumers shun chatbots in favor of human agents when chatbots' perceived capabilities are insufficient to complete the task—that is, when a chatbot is unable to do what is asked of it in the way in which it is asked (Luger and Sellen, 2016). Service-provider preference is also context- and individual-specific (Luger and Sellen, 2016; McLeay et al., 2021; Xu et al., 2020). For example, consumers avoid chatbot service in situations that carry perceived risk (Trivedi, 2019). Consumers who are less open to change, which is a form of risk aversion as a personality trait, also favor human service (McLeay et al., 2021).

In addition to risk, research identifies task complexity as an important situational factor that impacts preference for a service provider. For example, Xu et al. (2020) find that consumers prefer AI to help tackle low-complexity tasks and human service representatives—higher-complexity tasks. In their study, the effect is mediated by the agent's perceived problem-solving ability, lending further support for the aforementioned findings that highlight the importance of perceived AI capabilities (Luger and Sellen, 2016).

Of note, industry surveys consistently show that, at this point in time, consumers generally prefer to be served by human agents rather than chatbots. In one survey of 1000 US consumers, 80.5 percent of respondents expressed a preference for human agents and 19.5 percent of individuals preferred chatbots (Treasure Data, 2019). Consumers who preferred chatbots cited their higher efficiency and availability. Consumers who favored human service cited humans' better understanding of customer concerns and more thorough explanations, thus implicating empathy as a key factor. A separate international study of 6000 consumers (Pegasystems, 2019) also highlighted the importance of empathy in service interactions. In its most recent survey, Ipsos (2023) found that while only 68 percent of respondents in the survey panel had interacted with a chatbot, 88 percent of respondents preferred human customer service.

The overviewed research indicates that consumers will favor human agents over chatbots in service encounters that are non-routine in that they may entail problem solving, interaction complexity, outcome

uncertainty or risk. This literature motivates our first test of consumer responses to alternative service providers—a human versus a chatbot—following a non-routine service encounter. Formally.

H1. All else equal, individuals will report lower satisfaction, repatronage intentions, and more positive recommendation behavior following non-routine customer service interactions with a chatbot compared with a human agent.

For relevancy reasons, we use recommendation acceptance in the positive-outcome case and recommending to others in the negatively-valenced case.

2.2. Service outcome valence

Individuals are goal-directed agents whose actions, such as customer service encounters, seek to achieve particular outcomes (e.g., Ajzen, 1991). To the extent that service outcomes are judged based on one's goal attainment—success or failure—they embed positive or negative valence. Research shows that outcome valence impacts consumers' perceived experience with the service process, global service evaluations, and satisfaction (Heitmann et al., 2007). In their treatment of service encounters, Brady et al. (2006) conceptualize service outcomes as part of a service's "technical quality," because outcomes constitute part of what is delivered. Their empirical results show that outcome valence is a key antecedent of customer satisfaction, on par with the impact of functional quality and service environment quality. Empirical studies in various contexts, for example, sports and recreation, also support the outcome-service satisfaction link. The latter research stream shows that the outcomes of sporting events in which individuals are participants or spectators (with clear preferences), impact event evaluations and overall satisfaction (Clemes et al., 2011).

2.3. Emotions and empathy

The core human phenomena of emotions and empathy help elucidate anticipated differences in consumers' response to human and chatbot service-givers. Individuals experience a range of emotions in everyday life. These spill over and affect consumer behavior across various contexts (e.g., Frijda, 2007; Yu et al., 2024). Additionally, theories of human emotions indicate that emotions play a central role in goal-directed behavior (e.g., Frijda, 2007; Maduku, 2024). For example, the communicative theory of emotions (Oatley and Johnson-Laird, 1987) posits that emotions facilitate communication within the brain and among individuals.

Internally, emotions operate as signals that reflect appraisals of situations—are things good or bad? Are things going in the right direction? These signals play an evolutionary role by helping individuals to function in a complex world with cognitive resource constraints. For example, emotions help to manage one's effort in the presence of multiple goals by motivating focus on more important goals or based on proximity to goal attainment. Furthermore, individuals experience emotions when events or other people affect their progress towards goals. This helps to adjust goal-directed efforts. Accordingly, goal attainment and non-attainment produce emotions corresponding to the outcomes ranging from contentment and euphoria to anger, disappointment, and regret. It follows that complex feelings and emotions, or what Frijda (2007) called "emotion experiences," are pervasive in typically-goal-driven service situations.

Importantly, emotions play a key role in interpersonal communications and relationships informing others about our internal states, arousal, expectations, perceptions, attitudes, urges, and desires through our posture, gestures, facial expressions, voice, and language (Frijda, 2007). In that sense, consumers' emotions during customer service encounters transmit information. Service personnel must be able to sense, recognize, and respond appropriately to consumer emotions both for more effective communication and happy outcomes for consumers and

the organization (Hennig-Thurau et al., 2006).

However, human emotions are inherently difficult to read (Elfenbein and Ambady, 2002; Dogerlioglu-Demir et al., 2023). In service encounters, individuals may not always be direct in expressing how they feel or clearly articulate their position. There is also sometimes a disconnect between what people say they feel and what they actually feel (Aviezer et al., 2012). Moreover, emotion experiences are highly subjective—different people may experience a different mix of emotions in similar situations (Frijda, 2007). Variations in expressing and reading emotions are affected by gender, social status, and cultural peculiarities. For example, a smile may reflect different underlying emotions among the Americans, Japanese, and Thai; anger is more freely expressed in western cultures than Asian cultures unless one is of a high social status, when the opposite would occur (Park et al., 2013).

Human agents rely on their lifetime of experiences involving emotions in social interactions to help read and navigate customer emotions in service engagements. Their work-role performance also draws on the uniquely human ability to empathize with the customer (Pedersen, 2021; Wang et al., 2023). Scholars often distinguish between cognitive and affective empathy, and there is continued debate about how these constructs relate to each other (Cuff et al., 2016). Cognitive empathy refers to one's ability to take the position of and understand another person, whereas affective empathy is an "emotional response that stems from another's emotional state or condition and that is congruent with another's emotional state or condition (Eisenberg and Strayer, 1990, p. 5), or, as those authors put it, "feeling with another." Most empathy definitions incorporate both dimensions at the same hierarchical level (Cuff et al., 2016). Although, some authors (e.g., Strayer, 1987) argue that affective empathy represents the content of empathy while cognitive empathy is the process by which this content is produced (Cuff et al., 2016).

Being empathetic means caring about another's concerns and striving to understand what is being felt and thought. With empathy, one is better able to sense both spoken and unspoken emotions. One picks up subtle cues and listens attentively to understand the other person's point of view, the terms in which they think about the situation. In sum, a service agent's empathy can enable shared emotion experience with the customer, including the ability to see things from the customer's viewpoint, which enable better communication (Frijda, 2007; McBane, 1995).

Empathy in service personnel is highly valued by employers and customers (Delpechitre et al., 2018; McBane, 1995). A customer service agent's ability to recognize the various emotions and adjust accordingly—proceed at the appropriate pace, reassure, be 'human' and authentic—in other words, empathize enables the agent to establish trust, better resolve customer issues and, in the process, achieve higher customer satisfaction with the service experience (McBane, 1995; Prentice, 2019). For example, in healthcare, individuals are more likely to follow their medical practitioner's instructions and achieve better clinical outcomes if they feel heard and understood (James, 2023). Of note, empathy may be particularly important for remote customer sales and service when service agents do not get the benefit of observing customer body language or vocal expression (Li et al., 2023). The importance of empathy in services is formalized by treating empathy as a distinct dimension in the widely used SERVQUAL model (Parasuraman et al., 1985).

Although, when it comes to emotions, even the most advanced chatbots are at a disadvantage compared to human agents. Because of ubiquitous contextual complexities and nuances when emotions are displayed, for example, cultural influences, ethics, and social norms, such experience is difficult to develop in chatbot training (e.g., Agarwal, 2022). Chatbot capability arises from sophisticated algorithms trained on vast datasets. Generative AI uses patterns observed in training data to generate new content, including interpreting and responding to customer emotions. This process is based on probabilities and pattern matching rather than real emotion, human thought, empathy, etiquette,

and social norms. Currently, large language models (LLMs) underpinning generative AI programs can lose accuracy when presented with problems that feature altered or less common patterns. A recent paper testing LLM reasoning ability concludes: "Shortcut learning via pattern-matching may yield fast correct answers when similar compositional patterns are available during training but does not allow for robust generalization to uncommon or complex examples" (Dziri et al., 2023, p. 2). The quote refers to basic reasoning. Human emotions often present a more ambiguous challenge. Furthermore, chatbots have limited capacity to self-regulate or appropriately weigh the contextual (relative) importance of decision inputs when they process human emotions. These represent additional reasons why things can go wrong in service contexts (Beck, 2023) and why consumers may be hesitant to embrace chatbot service.

2.4. Consumer attributions

Communication theories help identify chatbots' limited ability to process human emotions and empathize as major impediments to communication and customer outcomes (Hill et al., 2015; Abd-Alrazaq et al., 2019; Ruan and Mezei, 2022). We combine these insights with attribution theory (Kelley, 1967; Weiner, 1985) to elucidate why consumers may respond differently to different service giver types when faced with positive versus negative service outcomes. The theory describes how individuals use information in their environment to find reasons or causal explanations for certain behaviors and events. Several attribution models have been proposed under the broad attribution theory umbrella. Weiner (1985) posits that consumers form cognitive causal structures consisting of three components: (1) the locus of causality identifies the perceived cause of an event, which can be internal or external to the individual; (2) the causal factor's stability characterizes its degree of permanence; and, (3) controllability refers to the extent to which the individual can influence it. Kelley (1967) further distinguishes between dispositional and situational attribution. Dispositional attribution links the cause of behavior or event to the actor's internal characteristics, whereas situational attribution assigns causality to forces or conditions that are outside a person's control.

The attribution framework can be applied to envision causal structures that consumers may form to make sense of their service experiences. To begin with, not all encounters justify an involved cognitive assessment on consumers' part. Some service encounters unfold as expected or are routine. Since individuals are "cognitive misers," routine service events, such as a straightforward product return, may be filed in memory without a causal assessment. However, service encounters often deviate from routine. Individuals are more likely to apply attribution reasoning to events and outcomes that are at least somewhat unexpected or unfavorable (Weiner, 1985). For example, consumers will naturally seek to understand and develop a causal explanation for a service that was not rendered exactly as requested. Such common non-routine service outcomes constitute the general context of our investigation.

Regarding controllability and the locus of causality, we note that consumers tend to have limited control over service outcomes, and hence their attribution reasoning will anchor on external factors, in particular, the company and its service personnel. We further note that consumers are aware that service provision is governed by company policies and its service model. Accordingly, most causal attributions for positive or negative customer service experiences are assigned to companies (or brands) rather than individual employees (Swanson and Kelley, 2001). For example, when a consumer receives competent service from an Apple employee, for which Apple is famous, this reinforces her belief that Apple has great customer service, rather than associating the outcome with a particular agent who happened to handle the issue. When a customer is dissatisfied with customer service delivery or outcome, they tend to blame the company and may resolve not to repurchase the brand in the future rather than placing the blame with the service representative.

At the same time, it is known that service agents (including sales and support) have certain autonomy in carrying out company policies at the point of service. They are empowered to show initiative while managing most aspects of service provision and customer problem solving. In this process, customer communication is a critical task whereby service agents gain an understanding of customer needs and show empathy and a willingness to help. Hence, the locus of causality in explanations of service outcomes may extend to service representatives, albeit in a limited way. The causal attributions will vary depending on the perceived degree of control that service agents possess in each case (Hess et al., 2003).

In considering consumer explanations for positive and negative outcomes, it is reasonable to assume that consumers will view service representatives as being generally well-meaning—that service representatives act to achieve customer satisfaction where possible and do not make arbitrary decisions leading to undesirable outcomes. This central assumption yields that the locus of causality and credit will be ascribed relatively more to service representatives in conjunction with positive outcomes than negative outcomes. That is, consumers will view positive service outcomes as achieved, at least in part, owing to the service agent's (human or AI) understanding and effort to find a solution within or sometimes around company policies.

Importantly, the aforementioned communication theories suggest that chatbots may receive a different amount of credit than human agents for achieving the same positive outcome under the same service conditions. As discussed previously, communication is critical in the service context. Consumers know and appreciate that human service agents possess greater ability than chatbots to understand and empathize with customers (van Doorn et al., 2017; Giebelhausen et al., 2014). Human agents will, thus, be perceived as being both more willing and able to help. Hence, they will be seen as more instrumental in successful service interactions. This perception will translate into more positive customer evaluations of service experience when accompanied by a positive outcome when served by a human agent. We express this formally as follows.

H2. After experiencing a positive service outcome, individuals will report higher satisfaction, repatronage intentions, and recommendation acceptance following interactions with a human agent compared with a chatbot.

In contrast to the predictions involving positive service outcomes, the agent benevolence assumption suggests a different causal attribution structure for negative outcomes that are not due to service failure. If consumers perceive service representatives as making a genuine effort to address their problem, failure to resolve the issue would be largely due to constraints imposed by company policies. Hence, individuals would view the company as the sole responsible party for a negative service outcome. Then, there should be no meaningful difference in consumer attributions of responsibility between chatbot and human agents. Accordingly, consumer evaluations of their service experience should be similar for both.

At the same time, an alternate causal attribution structure is more likely if consumers hold service agents at least somewhat responsible for negative service outcomes (Crolic et al., 2022). For example, a consumer may believe that a well-meaning agent, nevertheless, could have done more to help. It is also in the human nature to punish the bearer of bad news. Hence, service agents may be swept up in a vortex of negative emotions that tend to accompany service interactions culminating in negative outcomes.

Indeed, some empirical evidence on human-machine interactions indicates that consumers may tend to judge human representatives more harshly than chatbots when faced with a negative outcome. Consumers are more apt to believe that human agents have greater autonomy in their roles than machines (Choi et al., 2021). Humans certainly have a greater capacity for understanding and empathy (Pelau et al., 2021). Moreover, consumers are used to deriving higher value from service

interactions when served by humans and even extend this expectation to humanoid robots (Belanche et al., 2021). Hence, human agents are generally perceived to be in a better position to help than chatbots and may be more harshly judged when they fail.

A number of empirical studies provide indirect support for the main effect while highlighting various boundary conditions. Crolic et al. (2022) report five studies including one using real-world industry data to show that consumers attribute more blame to anthropomorphic chatbots (and the company) when they are angry. Choi et al. (2021) also find that consumers are more dissatisfied with humanoid service robots than nonhumanoid robots following a service failure, with the agent's perceived warmth mediating the effect. Pozharliev et al. (2023) compare consumer reactions to negative and positive outcomes involving autonomous vehicles (AVs) and vehicles with human operators. They find that consumers give the same credit to AVs and human agents for positive outcomes, but assign less blame to AVs for negative outcomes because the expect less from robotic vehicles. This leads us to the following hypothesis.

H3. After experiencing a negative service outcome, individuals will report higher satisfaction, repatronage intentions, and recommendation acceptance following interactions with a chatbot compared with a human agent.

2.5. Empathy as a mediator variable

As discussed previously, empathy facilitates human communication and interactions in social settings. In business settings, it helps sales and service agents to achieve better results for the company and its customers. In addition, several previous studies suggest that empathy may play a more central role in customer outcomes by mediating the effects of customer service efforts or service-giver intentions. For example, empathy and customer perspective-taking enable call center personnel to offer more effective customer support (Axtell et al., 2007). Customers' perceptions of salespersons' empathy mediate the relationship between salespersons' customer orientation and customer satisfaction (Gerlach et al., 2016). Also, consumer satisfaction with humanoid versus non-humanoid robot service is mediated by the robot's perceived warmth when consumers experience service process failure (Choi et al., 2021 operationalize warmth with empathy characteristics). In a similar vein, recent research using functional magnetic resonance imaging shows that empathy mediates individuals' preference to revisit a human medical provider over medical AI (Zhang et al., 2023). Research examining consumer acceptance of AI service robots also finds that the robots' perceived empathy mediates consumer trust and acceptance of such devices (Pelau et al., 2021). These findings suggest that the more positive impact of human agents on the focal customer response measures will flow through their higher perceived empathy. We anticipate observing the same mediation effect for both service provider types and outcomes. Stated formally.

H4. Perceived empathy mediates the effect of service-giver type (human or chatbot) on customer satisfaction, repatronage intentions, and recommendation acceptance following service interactions resulting in positive or negative outcomes.

We summarize the key relationships addressed in our hypotheses and the studies implemented to test them in Fig. 1.

3. Humanizing chatbot attributes to improve customer results

Since improving customer response to chatbot service is the primary motivation for this research, we next consider the service performance impact of three modifications to chatbots suggested by our theory and existing empirical evidence, with focus on the negative outcome case. There is considerable research exploring the perceptual and performance impacts of chatbot and robot anthropomorphism in the

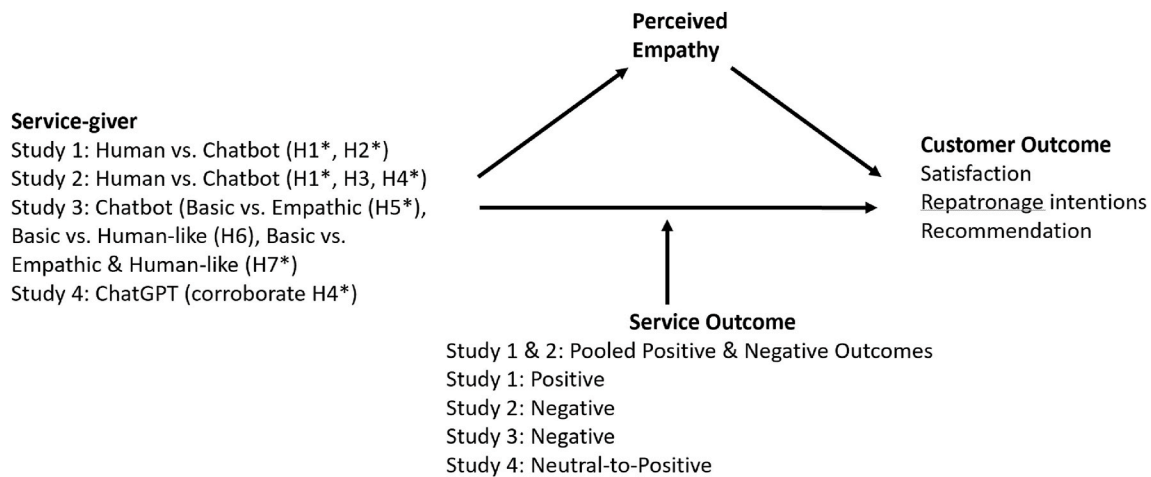


Fig. 1. General representation of the hypothesized relationships and hypotheses tested in each study. The asterisk indicates hypotheses that have been supported.

workplace and customer service contexts. Human-like features in chatbots that have been explored include a human name, gender, morphology (human body shape and limbs), human facial features or expressions, and human ways of communicating, including language that conveys caring, empathy or warmth.

Our central thesis in this research is that a service-giver's perceived empathy will play a key role in customer experience. Prior research on human-machine interactions specifically indicates that increasing a chatbot's perceived empathy during service provision can elevate consumer experience. For example, Liu and Sundar (2018) show that consumers respond more positively to health advice chatbots that display sympathy and affective empathy. Choi et al. (2021) explore recovery tactics following service process and service outcome failures when consumers are served by humanoid or non-humanoid robots. They observe a positive effect on customer satisfaction of increasing humanoid robots' perceived warmth and empathy via a sincere apology. Likewise, research in hospitality and tourism services shows that social robots' or chatbots perceived empathy affects individuals' intention to use the technology in hospitality settings (e.g., Orden-Mejia and Huerfano, 2022). Those findings indicate that consumers associate chatbot empathy with better service experience.

Therefore, we anticipate that increasing a service chatbot's perceived empathy relative to its less empathetic version would improve the focal customer metrics. Because visual and body language types of empathetic response to consumer requests in a chatbot service encounters are unavailable, we first consider empathetic verbal expression as an explicit display of empathy. We state our next hypothesis formally as follows.

H5. In the negative service outcome condition, individuals will report higher satisfaction, repatronage intentions, and recommendation likelihood after interaction with a chatbot that verbally communicates greater empathy compared with an equivalent chatbot with lower empathy.

Of note, we restrict this investigation to negative service outcomes because customer experience and emotions are depressed when customers fail to achieve their service objectives, and improvements in their service experience and perceptions may be of the greatest practical value at that time. We compare a more empathetic chatbot with a baseline chatbot rather than a human agent because we hypothesize (in H3) that a chatbot will outperform a human in the negative service outcome case. Hence, the baseline chatbot version is expected to present a higher performance bar than a human agent.

In contrast with the consistent results demonstrating that enhancing a chatbot service-givers' communicated empathy improves customer metrics, the literature is less conclusive on the effect of increased human

likeness, which may affect perceived empathy indirectly (Blut et al., 2021; Hsu and Lin, 2023). Namely, some authors show that increased human likeness is associated with elevated expectations of humanness, such as agency, warmth, empathy, and morality (e.g., Choi et al., 2021; Zhang and Wang, 2023). Other scholars argue that consumers ascribe human qualities to humanoid service providers. This tends to elicit a more positive reaction motivated by evolution-based social responses, similar to how one responds to humans (e.g., Duffy, 2003). As a result, consumers get more upset when humanoid chatbots fail to demonstrate human-like emotional states and understanding. Individuals do not display the same expectations and response toward nonhumanoid robots (Choi et al., 2021). Yet, some other scholars find that consumers may feel uncomfortable with increasingly humanoid robots because they perceive them as a threat to their human identity (Mende et al., 2019), elevating the likelihood of a more negative response to devices endowed with high human likeness.

On balance, multiple empirical studies report a generally beneficial impact of adding visual anthropomorphic features to robots or chatbots, in particular when such features are accompanied by verbal anthropomorphic cues (for a detailed overview, see meta-analysis by Blut et al., 2021). In their survey of individuals who previously interacted with a chatbot, Söderlund and Oikarinen (2021) find that chatbots' perceived humanness is positively associated with customer satisfaction. Brandtzaeg and Følstad (2018) report that consumers' negative reactions to highly humanoid robotic attendants is motivated by elevated expectations rather than consumer discomfort associated with the uncanny valley effect. Those authors and also Chong et al. (2021), who explore similar relationships, show that increasing a robot's perceived warmth (rather than competence) moderates the effect of human likeness on attitudes and behaviors toward robotic service. Pelau et al. (2021) show that anthropomorphic characteristics alone do not influence acceptance and trust towards robots. Instead, the effect flows through the robotic-provider's perceived empathy and interaction quality.

The reviewed research motivates our two final hypotheses pertaining to the impact of appearance (indirect) and verbal (direct) empathy attributes on consumer responses to chatbot service. Formally.

H6. In the negative service outcome condition, individuals will report higher satisfaction, repatronage intentions, and recommendation likelihood after interaction with a chatbot with greater visual human likeness compared with a chatbot with lower human likeness.

H7. In the negative service outcome condition, individuals will report higher satisfaction, repatronage intentions, and recommendation likelihood after interaction with a chatbot with greater human likeness and verbal empathy compared with a chatbot with lower human likeness and verbal empathy (the baseline case).

We summarize recent empirical research that explores individuals' reactions to AI-based service in [Table 1A](#) in Online Appendix A.

4. Research method

We conducted three experimental studies using the vignette method to explore the hypothesized effects. We chose to use the vignette method (e.g., [Huang et al., 2024](#); [Jörling et al., 2019](#)) for the initial studies due to the experimental control and internal validity that it allows. The method uses narratives to effect variable manipulation. Of note, vignettes have been used extensively in experimental research, including behavioral economics (e.g., [Kahneman and Tversky, 1979](#)). In consumer studies, this method has been shown to provide valuable insights into consumer thought processes and behaviors (e.g., [Grazzini et al., 2023](#)). Our fourth study, used to corroborate our findings concerning empathy, was a quasi-experiment in a field setting in which consumers interacted with ChatGPT to obtain medical advice. We note that quasi-experiments lack some experimental controls, but may compensate for that by enabling greater external validity.

4.1. Vignette studies participants and procedure

We used Amazon's MTurk and Prolific platforms to recruit our vignette study participants, including the pre-tests. We observed no demographic or measurement differences in our samples that were attributable to the different platforms. We conducted a power analysis to ensure adequate final samples. Participants were randomly assigned to seven experimental conditions across the three studies. All the studies opened with scales measuring experience with human service agents and virtual agents, including chatbots and virtual assistants online or on the phone. Next, we introduced our experimental vignettes (see Online Appendix B for details).

In Study 1 ($n = 199$), we instructed the participants to imagine that they were reserving their dream vacation. Their reservation was handled online either by a human (Amy Sullivan) or a chatbot (Stella). Participants were then shown a mock-up of their online chat with the reservation agent. The service interaction was non-routine by the virtue of its importance and the fact that consumers could not get their much-preferred tour guide and had to consider the reservation agent's recommendation for an alternative guide. Since the participants successfully made their reservation, the outcome was viewed as positive, which we verified in our pretests. We used this study to test [H2](#). We also combined Study 1 with Study 2 to test [H1](#) and [H4](#).

In Study 2 ($n = 200$), the participants were first asked to deliberate on their next smartphone purchase. They were then told to imagine that they bought the phone and intended to use it for some time, but it broke soon after the manufacturer warranty expired. (We collected a measure of the participants' phone replacement cycle and used it as a control in our robustness checks.) The participants were primed to feel negative emotions induced by the smartphone failure as they imagined contacting the manufacturer's customer service to ask that the device be repaired under warranty. They were informed that their request was handled by a human agent (Amy Sullivan) or a chatbot (Stella). Then they were presented with a mock-up of their online interaction which ended with their request being denied by the service agent. Our pretests confirmed that individuals viewed the outcome as negative.

Study 3 ($n = 315$) used the same negative-outcome smartphone repair setting as Study 2. It featured four chatbot conditions: low empathy/low human likeness (i.e., the baseline chatbot condition), increased empathy/low human likeness, low empathy/increased human likeness, and increased empathy/increased human likeness. The low empathy conditions featured the same interaction as in Study 2. Consistent with prior research that explored empathy in robot interactions (e.g., [Blut et al., 2021](#); [Choi et al., 2021](#); [Chong et al., 2021](#); [Pelau et al., 2021](#)), out increased-verbal-empathy conditions included a more empathetic tone, better explanation for the rejection, and a more

heart-felt apology.

We used multiple pretest studies to construct a chatbot with an effective increase in communicated empathy relative to the baseline chatbot. We also separately conducted pretests to select a favorable human likeness from multiple chatbot appearances ranging from a schematic outline to a fully human face created by generative AI. Our selected moderately human-like chatbot representation (see Online Appendix B) was ranked a close second to the fully humanized chatbot version on our pre-test measures of favorability and preference.

4.2. Constructed measures and controls

We took several steps to ensure data quality. In Study 1, we asked our participants whether they had travel reservation experience and whether those served by Stella understood that Stella was a chatbot. For Studies 2 and 3, we verified that individuals both had and intended to repurchase a smartphone. To increase participant involvement, the scenario referred to the phone the participants indicated they were planning to purchase as their next smartphone. The final samples by condition, after our manipulation and attention checks, are shown in the results tables.

All the response alternatives in our rating scales discussed below are measured based on a seven-point scale. [Table 1](#) shows the construction of our key variables and controls (described in the following sections), as well as the associated reliability coefficients.

To test customer satisfaction, we use a three-item measure, *Satisfaction Index*, which captures the extent to which participants were satisfied with the service interaction overall (phrased as "reservation interaction" in Study 1), satisfaction with the service agent specifically ("reservation agent" in Study 1), and satisfaction with the customer service experience provided by the firm. In addition, we use two measures of behavioral intent. We measure loyalty towards the provider by asking about the likelihood of using the same reservation agency (smartphone brand) on a future travel (purchase) occasion. This is a single-item measure that captures a distinct behavioral loyalty dimension—repatronage intentions (e.g., [Söderlund, 2006](#)). In Study 1, we measure recommendation acceptance by asking about the likelihood of calling back to modify the reservation so as to get one's preferred tour guide. In Studies 2 and 3, we measure the likelihood of recommending the smartphone brand to friends and family.

Our three-item measure of perceived empathy is adapted from the SERVQUAL scale, which captures empathy as one of the five distinct dimensions of service quality ([Parasuraman et al., 1985](#)). To account for possible confoundedness in our analyses, we control for the extent of recent interactions with (1) chatbots and AI assistants, and (2) human service agents.

5. Vignette study results

As the first step in our analyses, we tested for significant differences between experimental conditions on the demographic variables of age and sex. None of the mean differences are significant at $p < 0.1$. We also compared the conditions based on a separate check about the extent to which the participants were concerned about AI replacing human interaction and jobs, since strong feelings in this regard could bias our results. On average, our participants displayed low concern about AI, with mean responses ranging between 3.9 and 4.3 out of seven across the cells. We report descriptive statistics and correlations for our key variables in [Table 2](#).

5.1. Main analyses

We used ordinary least squares regression (OLS) to test our hypotheses by regressing our three dependent variables (DVs) on the service giver type dummy variable which took the value of one if the service giver was a chatbot and zero if human. All the regressions also

Table 1
Variable construction and construct reliability (coefficient alpha).

DVs	Items	Alpha
1. Satisfaction index	i. How satisfied or dissatisfied were you with your reservation (customer service) ^a interaction overall? ii. How satisfied or dissatisfied were you with your reservation (customer service) ^a representative? iii. Overall, customers would be happy with a customer service experience with a firm like this one (Strongly agree – Strongly disagree).	0.87
2. Repatronage intentions – safari reservation scenario	i. If you were to make a travel reservation in the future, how likely would you be to use the same reservation agency/tour operator?	
Repatronage intentions – smartphone repair scenario	i. If you were to buy a phone in the future, how likely would you be to use the same brand from the scenario you imagined? ^a	
3. Recommendation acceptance	ii. How likely would you be to call again and modify your reservation to get your initially chosen tour guide, Jackson Looseyia? ^b	
Recommending ^a	i. Based on the service you received today, how likely are you to recommend this phone brand to your friends and family? ^a	
IVs		
Perceived empathy (based on SERVQUAL)	I think that the Customer Service ... i. Has the customers' best interests at its core. ii. Provides individualized attention. i. Understands the needs of their customers.	0.90
Controls		
1. Experience w/chatbots	i. In the past 2 years, how much experience did you have interacting with Virtual Assistants (i.e. Siri, Alexa, Google Assistant)? ii. In the past 2 years, how much experience did you have interacting with customer service Chatbots online?	0.83
2. Experience w/human agents	i. In the past 2 years, how much experience did you have interacting with human customer service representatives online? ii. In the past 2 years, how much experience did you have interacting with human customer service representatives over the phone?	0.82

Notes: All the items were measured on a 1–7 scale.

^a Indicates the version of the measure (modified language) that was used in Studies 2 and 3.

^b The item was reverse-scored.

included two control variables—the extent of experience with chatbots and human customer service. We began our main analysis by pooling the data from Study 1 and Study 2 to test for individuals' prevalent response to chatbot versus human service across our conditions. The results are shown in Table 3. All the regressions are significant at $p < 0.05$ or better. The coefficient on the Chatbot service dummy is highly significant (at $p < 0.01$) and negative for all three DVs. This set of results supports H1 which posits that, all else equal, individuals, on average, will respond more positively to human service than chatbot service for all outcomes.

Next, we separately evaluated the data generated by Study 1 and Study 2 to get a more fine-grained understanding of consumer responses to service-giver type as a function of service outcomes. Consistent with hypothesis H2, the coefficients on the chatbot service dummy are all significantly negative at $p < 0.05$. Therefore, we find that individuals respond significantly more positively to human service than chatbot service when they achieve a positive outcome. Contrary to H3, the results are similar in the negative outcome condition, thus buttressing the

alternative conclusion that consumers currently favor human agents in non-routine service encounters even when they are not happy about the outcome. We show our tests of H1–H3 in Panel A of Table 3.

To test our mediation hypothesis H4, we ran mediation analyses for each DV. First, we evaluated our models with perceived empathy included as a regressor. The results are shown in Table 3, Panel B. We observe that the coefficient on perceived empathy is significantly positive at $p < 0.01$ while the coefficient on chatbot service dummy lost significance in all the regressions but one—for customer satisfaction in the positive outcome condition. We next used Hayes (2018) PROCESS macro (model 4) with 5000 bootstrap samples. We specified the chatbot service dummy as the independent variable and perceived empathy as the mediator. This analysis produced significant (at $p < 0.05$) coefficients on the indirect effect of perceived empathy in all the regressions. We summarize our mediation analysis results in Table 4. In sum, our data provide full support for H4 which states that perceived empathy mediates the effect of service-giver type on consumer evaluations of their service experience when faced with positive or negative service outcomes.

Finally, Study 3 evaluated three practical approaches to increasing the favorability of chatbot service evaluations in order to improve chatbot acceptance by consumers. The first approach entailed increasing a chatbot service-giver's perceived empathy by providing a better (more considerate) explanation for the negative decision, expressing understanding of the customer's feelings, and offering an authentic apology. The second approach included a more human-like representation of the chatbot. The third approach combined the two. To test our hypotheses, we regressed our dependent variables on the dummy variable labeled "Condition dummy" which took the value of one for the tested experimental condition and zero for the baseline condition featuring a basic chatbot graphic and lower empathy. The regressions were fit only in the negative outcome condition. We show the results in Table 5. The levels of significance in this table are based on one-sided tests, since we are testing directional hypotheses.

Our tests of hypotheses H5–H7 produced marginally significant results for some DVs in support of H5 and H7, but not H6. The observed positive coefficients on the experimental condition dummy reveal that individuals responded more positively to chatbots that were more empathetic during the service interaction. Our participants showed no increase in evaluation favorability when served by a more anthropomorphic-looking chatbot when it did not express increased empathy.

5.2. Sensitivity analyses

We verified that including age and sex as additional controls did not affect our results. We also evaluated a fully human-like chatbot as an additional condition in our sensitivity analyses in Study 3, since this chatbot representation received directionally higher favorability scores in our pretests. In this sensitivity analysis, the fully human-like chatbot included increased verbal empathy. Surprisingly, it received marginally lower scores on our DVs than our primary chatbot configuration with moderate human likeness and increased empathy (as tested in H7).

Since we observed that consumers preferred a human agent even after experiencing a negative service outcome, we evaluated an additional model which compared a human agent in the baseline condition (with low empathy) against our best-performing chatbot implementation which featured increased empathy. We show these results in the three right-most columns in Table 5. None of the regressions are significant, indicating that there were no meaningful differences between an empathetic chatbot and a human agent on our measures of customer satisfaction and behavioral response.

6. Study 4 – quasi-experiment

Although the vignette-based method offers important

Table 2

Descriptive statistics and correlations in our seven experimental cells in Studies 1–3.

Service outcome conditions:	#	Variable	Mean	S.D.	1	2	3	4	5
<i>Panel A: Positive outcome</i>									
Human agent	1	Satisfaction Index	5.479	0.908	1				
/base empathy	2	Provider Loyalty	5.295	1.367	0.746*	1			
N = 95	3	Recommend. Accept	3.895	1.888	0.114	0.136	1		
	4	Perceived Empathy	5.526	1.115	0.630*	0.653*	−0.108	1	
	5	Exper. w/chatbots	4.405	1.260	−0.125	−0.055	−0.232*	−0.003	1
	6	Exper. w/human serv.	4.595	1.325	0.039	0.058	−0.187	0.166	0.262*
Chatbot	1	Satisfaction Index	4.788	1.363	1				
/non-humanized	2	Provider Loyalty	4.625	1.638	0.858*	1			
/base empathy	3	Recommend. Accept	3.394	1.803	0.319*	0.386*	1		
N = 104	4	Perceived Empathy	4.846	1.435	0.694*	0.614*	0.350*	1	
	5	Exper. w/chatbots	4.346	1.364	0.313*	0.15	−0.009	0.290*	1
	6	Exper. w/human serv.	4.495	1.480	0.242*	0.135	−0.012	0.268*	0.571*
<i>Panel B: Negative outcome</i>									
Human agent	1	Satisfaction Index	3.965	1.588	1				
/base empathy	2	Provider Loyalty	3.635	1.825	0.509*	1			
N = 96	3	Recommending	3.292	1.710	0.603*	0.814*	1		
	4	Perceived Empathy	4.260	1.549	0.623*	0.429*	0.523*	1	
	5	Exper. w/chatbots	4.083	1.291	−0.078	−0.002	0.005	−0.033	1
	6	Exper. w/human serv.	4.427	1.527	−0.065	−0.002	0.002	−0.086	0.530*
Chatbot	1	Satisfaction Index	3.346	1.455	1				
/non-humanized	2	Provider Loyalty	3.000	1.678	0.549*	1			
/base empathy	3	Recommending	2.731	1.614	0.659*	0.792*	1		
N = 104	4	Perceived Empathy	3.253	1.627	0.687*	0.434*	0.510*	1	
	5	Exper. w/chatbots	4.035	1.328	0.091	0.07	0.106	0.136	1
	6	Exper. w/human serv.	4.385	1.430	0.051	−0.087	−0.016	0.063	0.484*
Chatbot	1	Satisfaction_index	3.835	1.545	1				
/non-humanized	2	Provider Loyalty	3.406	1.924	0.559*	1			
/high empathy	3	Recommending	3.208	1.91	0.614*	0.880*	1		
N = 101	4	Service_Empathy	3.934	1.672	0.775*	0.601*	0.613*	1	
	5	Exper. w/chatbot	4.069	1.451	0.162	0.174	0.166	0.193	1
	6	Exper. w/human-agents	4.455	1.602	0.061	0.098	0.106	0.041	0.374*
Chatbot	1	Satisfaction Index	3.788	1.641	1				
/humanized	2	Provider Loyalty	3.159	1.874	0.418*	1			
/base empathy	3	Recommending	2.953	1.745	0.437*	0.798*	1		
N = 107	4	Perceived Empathy	3.623	1.752	0.738*	0.434*	0.499*	1	
	5	Exper. w/chatbots	3.925	1.338	−0.093	−0.017	0.021	0.151	1
	6	Exper. w/human serv.	4.519	1.426	−0.119	−0.086	−0.049	0.074	0.473*
Chatbot	1	Satisfaction Index	3.732	1.594	1				
/humanized	2	Provider Loyalty	3.523	1.855	0.567*	1			
/high empathy	3	Recommending	3.280	1.816	0.626*	0.872*	1		
N = 107	4	Perceived Empathy	3.726	1.769	0.768*	0.611*	0.688*	1	
	5	Exper. w/chatbots	4.003	1.259	−0.135	−0.006	0.019	−0.125	1
	6	Exper. w/human serv.	4.570	1.479	0.031	−0.084	0.007	−0.083	0.232*

Notes: S.D. is standard deviation. N is the number of valid observations. * denotes statistical significance at $p < 0.05$.

methodological advantages by enabling experimental control, it relies on the imaginative capacity of the participants, which may limit the method's external validity. To overcome this challenge, we conducted Study 4 as a quasi-experiment in which participants interacted with an actual AI chatbot—ChatGPT—in a manner that reflects how consumers use generative AI in practice (i.e. Kirshner, 2024). The aim of our Study 4 was to corroborate the role of empathy in customer service satisfaction both in a real world setting and a new context relative to our travel reservation and product repair service settings. In terms of outcome valence, successfully receiving medical advice from an AI chatbot may be qualified as a positive-to-neutral outcome, since the participants could achieve their objective with one or more prompt iterations in what was a low-stakes service interaction.

6.1. Participants, procedure, and measures

Cloud Research's Connect platform was used to recruit 100 (50% female) participants for this study. All participants were required to have access to ChatGPT. Participants were instructed to visualize experiencing flu-like symptoms and ask ChatGPT for advice. We used prompt guidelines and a sample prompt to help generate ChatGPT responses that varied in perceived empathy while limiting the response length to 200 words (in order to increase participant interactions with

ChatGPT output). We pre-tested our instruments to ensure their desired effects.

We randomly offered participants one of two prompt versions which elicited either more formal or more empathetic tone in ChatGPT's responses. In the spirit of a field experiment, participants had the freedom to construct their own prompt based on our guidelines or use our sample prompt. Participants were allowed to run as many prompt iterations as necessary to obtain what they felt was an adequate piece of advice for someone experiencing flu-like symptoms. Because we had limited control over participants' final prompt and ChatGPT responses we did not seek to construct high- and low-empathy experimental conditions. Instead, our manipulation objective was to induce variation in ChatGPT's perceived empathy across our study participants. In conclusion of the study, participants uploaded their final prompt and ChatGPT's response and then responded to our dependent variable measures and demographic questions. We show our instruments and prompt details in Online Appendix C.

Study 4 used the same variables and controls as Studies 1–3. In addition, we measured participants' experience with generative AI specifically and the participants' concern that AI was replacing human interaction and jobs. We used the latter as an additional control.

Table 3

Regressions of customer response variables on service-giver type (chatbot or human) and controls. These analyses are based on Study 1 and 2.

Panel A: Main Model	Test of H1 : Pooled Data			Test of H2 : Positive Outcome Conditions			Test of H3 : Negative Outcome Conditions		
	Satisfaction	Repatronage	Recommend	Satisfaction	Repatronage	Recommend	Satisfaction	Repatronage	Recommend
Chatbot service	−0.644*** (0.154)	−0.644*** (0.184)	−0.534** (0.178)	−0.675*** (0.164)	−0.658** (0.215)	−0.515* (0.261)	−0.620** (0.216)	−0.636* (0.248)	−0.561* −0.236
Experience/w chatbot service	0.019 (0.065)	0.033 (0.077)	−0.063 (0.075)	0.087 (0.069)	0.028 (0.091)	−0.126 (0.111)	−0.012 (0.090)	0.075 (0.103)	0.001 (0.098)
Experience/w human service	0.078 (0.061)	0.039 (0.073)	−0.022 (0.071)	0.105 (0.065)	0.100 (0.085)	−0.070 (0.103)	−0.003 (0.086)	−0.088 (0.098)	−0.008 (0.094)
Constant	4.284*** (0.307)	4.141*** (0.365)	3.969*** (0.355)	4.614*** −0.341	4.710*** −0.448	4.771*** −0.545	4.031*** (0.412)	3.696*** (0.473)	3.322*** (0.450)
adj. R2	0.042	0.025	0.018	0.101	0.043	0.018	0.026	0.022	0.013
F	6.817	4.457	3.392	8.396	3.979	2.224	2.743	2.481	1.881
p	0.000	0.004	0.018	0.000	0.009	0.087	0.044	0.062	0.134
df_m	3	3	3	3	3	3	3	3	3
N	399	399	399	199	199	199	200	200	200
Panel B: Mediation Model	Test of H1 : Pooled Data			Test of H2 : Positive Outcome Conditions			Test of H3 : Negative Outcome Conditions		
	Satisfaction	Repatronage	Recommend	Satisfaction	Repatronage	Recommend	Satisfaction	Repatronage	Recommend
Chatbot service	−0.065 (0.109)	−0.0697 (0.151)	−0.178 (0.170)	−0.280* (0.128)	−0.164 (0.174)	−0.335 (0.266)	0.0104 (0.172)	−0.16 (0.236)	−0.0183 (0.213)
Experience/w chatbot service	−0.004 (0.044)	0.010 (0.061)	−0.077 (0.069)	0.033 (0.053)	−0.039 (0.072)	−0.151 (0.110)	−0.022 (0.068)	0.067 (0.093)	−0.008 (0.084)
Experience/w human service	0.007 (0.042)	−0.032 (0.058)	−0.066 (0.065)	0.001 (0.050)	−0.029 (0.068)	−0.117 (0.103)	0.008 (0.065)	−0.079 (0.089)	0.002 (0.080)
Empathy	0.697*** (0.033)	0.690*** (0.045)	0.428*** (0.051)	0.601*** −0.05	0.751*** −0.067	0.274** −0.103	0.626*** (0.052)	0.473*** (0.071)	0.539*** (0.064)
Constant	1.299*** (0.253)	1.184*** (0.349)	2.136*** (0.394)	2.007*** −0.336	1.453** −0.457	3.581*** −0.698	1.361*** (0.382)	1.679** (0.524)	1.021* (0.473)
adj. R2	0.55	0.384	0.163	0.486	0.414	0.048	0.44	0.199	0.272
F	122.6	62.99	20.39	47.83	35.91	3.495	40.07	13.36	19.6
p	0.000	0.000	0.000	0.000	0.000	0.009	0.000	0.000	0.000
df_m	4	4	4	4	4	4	4	4	4
N	399	399	399	199	199	199	200	200	200

Notes: Standard errors are in parentheses.

*p < 0.05, **p < 0.01, ***p < 0.001.

Table 4

Summary of mediation analysis tests based on Hayes (2018) PROCESS macro with 5000 bootstrap samples. These analyses are based on Study 1 and 2.

Independent Variable	Mediator	Dependent Variable	b	95% CI	Significance
<i>Panel A: tests of H4 in the positive outcome condition</i>					
Service-Giver Type	Perceived Empathy	Satisfaction Index	−0.413	[−0.6503, −0.1954]	p < 0.05
Service-Giver Type	Perceived Empathy	Repatronage Intention	−0.050	[−0.7865, −0.2342]	p < 0.05
Service-Giver Type	Perceived Empathy	Recommend. Accept.	−0.149	[−0.3481, −0.0017]	p < 0.05
<i>Panel B: tests of H4 in the negative outcome condition</i>					
Service-Giver Type	Perceived Empathy	Satisfaction Index	−0.203	[−0.3164, −0.0852]	p < 0.05
Service-Giver Type	Perceived Empathy	Repatronage Intention	−0.163	[−0.2585, −0.0677]	p < 0.05
Service-Giver Type	Perceived Empathy	Recommending	−0.180	[−0.2785, −0.0748]	p < 0.05

Notes: b is the coefficient on the indirect effect of the mediator variable on the dependent variable.

CI is the confidence interval.

6.2. Analysis and results

We used OLS to test our hypotheses by regressing our focal dependent variables on the empathy variable, prior experience with generative AI, and controls. The regression results are shown in Table 6. We found that, consistent with our results in Studies 1–3, ChatGPT's perceived empathy significantly positively affected user satisfaction (all p's < 0.001). We also found that previous experience with generative AI did not influence satisfaction when perceived empathy was included in the model. We then further explored the role of service provider's perceived empathy in customer satisfaction by running a mediation analysis, similar to Study 3. Since this study did not use a human agent to contrast with ChatGPT, we could not replicate the mediation set-up from Study 3. Instead, we used prior experience with generative AI as the independent variable (e.g., Jan et al., 2023; Li & Wang) and empathy as

a mediator. Once again, we found that the effect of experience on satisfaction and recommending behavior was significantly mediated by empathy (all p's < 0.05) (see Table 7).

Study 4 generated several unique insights in combination with the first three studies. First, this study entailed a quasi-experiment involving an interaction with an AI chatbot that is in widespread public use. This allowed us to replicate and extend our findings from studies 1–3 on empathy in a real world situation. Second, this study explored additional control measures to help rule out alternative explanations. Specifically, we found a significant impact of experience on satisfaction. Importantly, though, from a process perspective, this effect was mediated by empathy, providing further support that perceived empathy is a key factor in consumer satisfaction with chatbot service.

Table 5
Regressions of customer response variables on chatbot service-provider. These analyses are based on Study 3.

Chatbot conditions	H5: Low human likeness, High empathy			H6: High human likeness, Low empathy			H7: High human likeness, High empathy			Low human likeness, High empathy		
	vs. Low human likeness, Low empathy	vs. Low human likeness, High empathy	vs. Low human likeness, Low empathy	vs. Low human likeness, Low empathy	vs. Low human likeness, Low empathy	vs. Low human likeness, Low empathy	vs. Low human likeness, High empathy	vs. Low human likeness, High empathy	vs. Low human likeness, High empathy	vs. Human agent	vs. Human agent	vs. Human agent
DV:												
Condition	Satisfaction	Repatronage	Recommend	Satisfaction	Repatronage	Recommend	Satisfaction	Repatronage	Recommend	Satisfaction	Repatronage	Recommend
dummy	0.491** (0.210)	0.418* (0.251)	0.482* (0.247)	0.425* (0.216)	0.203 (0.247)	0.243 (0.235)	0.368* (0.211)	0.553** (0.244)	0.556** (0.239)	-0.124 (0.224)	-0.218 (0.268)	-0.0768 (0.259)
Experience/w	0.079 (0.081)	0.184+ (0.097)	0.122 (0.096)	-0.074 (0.089)	0.098 (0.101)	0.050 (0.096)	-0.082 (0.084)	0.082 (0.097)	0.045 (0.095)	0.067 (0.087)	0.129 (0.104)	0.088 (0.101)
chatbot service	0.022 (0.078)	-0.061 (0.093)	0.009 (0.092)	-0.009 (0.086)	-0.153 (0.098)	-0.062 (0.093)	0.073 (0.079)	-0.135 (0.092)	-0.021 (0.090)	-0.030 (0.082)	0.004 (0.098)	0.029 (0.095)
human service	2.909* (0.392)	2.469* (0.469)	2.165* (0.461)	3.706* (0.417)	3.247* (0.476)	2.788* (0.452)	3.381* (0.418)	3.236* (0.484)	2.630* (0.472)	3.804*** (0.413)	3.051*** (0.493)	2.775*** (0.478)
Constant	0.019	0.016	0.014	0.011	0	-0.008	0.008	0.018	0.012	-0.011	-0.001	-0.007
adj. R2	2.338	2.075	1.975	1.761	0.972	0.47	1.547	2.282	1.855	0.308	0.934	0.519
F	0.075	0.105	0.119	0.156	0.407	0.703	0.204	0.080	0.138	0.820	0.425	0.670
p	205	205	205	211	211	211	211	211	211	197	197	197
N												

Notes: The Condition dummy takes the value of one for the tested condition listed first in the test title. The dummy took the value of zero for the chatbot with low human likeness and empathy.

Standard errors are in parentheses. Hypotheses H5-H7 are directional and were tested using one-sided significance testing.

*p < 0.05, **p < 0.01.

Table 6
Regression of customer response variables in a task involving ChatGPT. These analyses are based on Study 4.

IV \ DV	Satisfaction with Chatbot	Satisfaction with Advice	Recommend
Perceived Empathy	0.505*** (0.065)	0.503*** (0.069)	0.623*** (0.076)
Previous Experience	-0.013 (0.055)	-0.011 (0.059)	0.137 (0.065)
with Generative AI	-0.014 (0.048)	0.017 (0.051)	0.018 (0.056)
Concern that AI is			
Replacing Human			
Interactions and Jobs			
Age	0.001 (0.000)	0.000 (0.000)	0.000 (0.001)
Gender	0.207 (0.158)	0.314 (0.168)	0.334 (0.186)
Constant	2.817*** (0.495)	2.582*** (0.527)	1.208*** (0.580)
adj. R ²	0.39	0.368	0.471
F	13.782	12.65	18.781
p	0.000	0.000	0.000
N	100	100	100

7. Discussion and conclusions

7.1. Summary of main results

Our findings show that consumers reported significantly lower satisfaction, repatronage intentions, recommendation acceptance, and recommending the provider to others following interactions with a chatbot compared with a human agent in both positive and negative service outcome conditions. The effect was fully mediated by the service-giver's perceived empathy. Increasing a chatbot's perceived empathy via more empathetic communication (but not human-like appearance) improved consumer evaluations of chatbot service relative to a less empathetic chatbot configuration. Further tests show no significant difference on the focal measures between a more empathetic chatbot and a human agent in the baseline empathy condition, suggesting that the tested chatbot configuration demonstrated non-inferiority with human service. We obtained corroborating evidence concerning the impact of a service-giver's perceived empathy on consumer satisfaction and empathy's mediating role via a quasi-experiment involving study participants' interactions with an actual generative AI chatbot, ChatGPT.

7.2. Contributions

Considerable scholarly and practitioner efforts are directed toward understanding the factors and boundary conditions that impact consumer acceptance of chatbot service. Our results address a notable gap in this literature. Industry surveys reveal persistent consumer preference for human service over chatbots. However, the studies present consumer responses in the aggregate, which averages out variation in consumer experience and evaluations induced by situational factors. One ubiquitous factor involves service outcomes themselves, since consumers are known to respond differently to differently-valenced outcomes (Brady et al., 2006; Jörling et al., 2019).

Our research enables a more fine-grained understanding of consumer reactions to chatbot service when consumers experience positive versus negative service outcomes in non-routine service encounters that prompt consumer sense-making and attribution reasoning (Weiner, 1985). We show that consumers report less favorable reactions to a chatbot than a human agent following both positive and negative service outcomes. While the former result is consistent with the current empirical evidence that consumers generally prefer human service, the latter result runs counter to arguments that consumers may feel better about service that had a negative outcome if they were served by a chatbot. This reasoning holds that consumers expect less from

Table 7
Mediation analysis results based on Study 4 data capturing participants' interactions with ChatGPT.

Independent Variable	Mediator	Dependent Variable	b	95% CI	Significance
Previous Experience with Gen AI	Perceived Empathy	Satisfaction with ChatGPT	0.119	[0.0384, 0.2133]	$p < 0.05$
Previous Experience with Gene AI	Perceived Empathy	Satisfaction with Advice	0.071	[0.2579, 0.5344]	$p < 0.01$
Previous Experience with Gene AI	Perceived Empathy	Recommending	0.159	[0.0216, 0.2966]	$p < 0.03$

non-humanoid automated attendants and, hence, may apportion less blame to the service-giver in the face of dashed expectations (Belanche et al., 2021; Crolic et al., 2022; Grazzini et al., 2023; Pozharliev et al., 2023). Our results for the mediating role of empathy suggest a likely explanation for our contrasting results—that individuals treat provider empathy as integral to their service experience and respond generally more negatively when they perceive lack thereof.

Our finding that perceived empathy mediates the effect of service provider type on customer response measures in all outcome conditions represents our second contribution. It elucidates an important mechanism that links the service-giver with the customer's service experience. It adds to the body of knowledge that reinforces the central role of empathy in customer encounters involving not only human agents, but also chatbot service providers (e.g., Blut et al., 2021; Prentice, 2019; Pedersen, 2021).

Building on our findings concerning the central role of empathy, our third contribution consists in generating nuanced empirical evidence on the relative effectiveness of three approaches to enhancing a chatbot service-giver's perceived empathy in order to improve customer experience following a negative service outcome. We show that a direct verbal expression of empathy during a service interaction that ends on a low note is effective. This result aligns with past research findings in several studies, in particular, by Grazzini et al. (2023) who showed that consumers responded positively to social robots when they communicated warmth rather than competence. Our results for the impact of humanoid appearance in a chatbot service-giver, however, support the view that find this aspect of chatbot entity less important (e.g., Li and Zhang, 2023; Chi and Hoang Vu, 2023; Pelau et al., 2021). We believe that the subdued impact of appearance may be due to the increased importance of text in a chat interaction such as ours, when individuals' attention is fixed on the verbal exchange. By contrast, many prior findings supporting the beneficial role of automated attendant appearance were generated in conjunction with robots which possess a physical presence (e.g., Belanche et al., 2021; Blut et al., 2021; Grazzini et al., 2023).

7.3. Managerial implications

Chatbots have become an essential technology component of an organization's customer service strategy. The increased automation is meant to drive improvements in service efficiency and, eventually, effectiveness. Although, customer-oriented organizations will seek to roll out chatbot service without sacrificing their marketing objectives, including customer satisfaction and loyalty behavior.

Our research findings enable us to make several recommendations concerning steps that managers can take (or avoid) in order to improve chatbot performance on customer service measures. First, we find no difference in consumer response to chatbot service as a function of service outcome valence—consumers, generally, respond more favorably to human service following both positive and negative service outcomes. Hence, conditioning service-giver type on the likely service outcome based on the information collected prior to a service encounter, while feasible, represents a dead end. This is useful to note in light of the ongoing effort to optimize technology deployment.

By contrast, increasing a chatbot's ability to communicate empathy during a service encounter may prove beneficial. Our tests in the negative outcome condition indicate that programming a chatbot for enhanced verbal empathetic expression can markedly improve

consumer response to chatbot service. Recognizing that artificial empathy is a long-term goal (Raman and Flynn, 2024), organizations may achieve initial success by enabling their chatbot to provide a better, more considerate explanation for the decision or to issue a deeper, more authentic apology, as appropriate. Of note, we did not find that endowing a chatbot with a more humanoid appearance achieved a similarly beneficial impact on service evaluations. Our additional comparisons of a more empathetic chatbot versus a human agent revealed that customer satisfaction with the empathetic chatbot service approached the levels of consumer response to human service. This suggests that empathetic chatbots may successfully aid consumers in challenging service contexts and, thus, enable full automation of service provision.

7.4. Limitations and future research

We focus on non-routine service interactions that involve a customer service agent's notable intervention and a deviation from the expected, which prompts subsequent consumer sense-making (Weiner, 1985). Our results reveal persistent preference for human service. Consumers may respond differently to service-givers in situations that are routine, and in which a chatbot or a human agent performs a mechanical function—for example, as in a basic product return. Just as individuals respond positively to basic service automation, consumers may embrace chatbots in such roles in contrast to human or even menu-driven service. Research into chatbot reception in routine service situations may uncover managerially relevant insights.

In addition, our experimental setting charts a streamlined service interaction, with no wait, service delays, or opportunities to select a chatbot or human agent for service provision. We believe it would be instructive to examine whether consumer responses can be influenced by the distinct paths that consumers may take leading up to and through their service encounter.

Further, we contextualize this research in diverse customer service applications that entail service provided by human and chatbot agents via online chat. The degree to which our recommendations may extend to face-to-face service contexts (e.g., involving humanoid robots), or voice-based interactions, which, both, enable a distinct set of cues and limitations would need to be established in future research. By extension, research that explores how adjusting chatbot expression or appearance in response to observable customer cues may impact customer outcomes (e.g., see van Pinxteren et al., 2023 research on mimicry) may help to further inform ways of increasing consumer acceptance of chatbot service.

CRediT authorship contribution statement

Dmitri G. Markovitch: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rusty A. Stough:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dongling Huang:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – review & editing.

Declaration of competing interest

None

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jretconser.2024.103847>.

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